

Factors affecting transfer of online training: A systematic literature review and proposed taxonomy

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Abstract

Training employees in new competencies is crucial for ensuring organizational sustainability and success. Consequently, organizations allocate significant resources to training initiatives. As training constitutes a substantial investment, it becomes imperative to identify and categorize factors that enhance the transferability of training. This necessity becomes particularly pronounced in the context of online training, which is being increasingly adopted by organizations for various reasons, including cost efficiency and flexibility. While the adoption of online training is on the rise, research on factors that enhance the effectiveness of such training remains limited. This study aims to address this gap by introducing a taxonomy of factors that influence the transfer of online training. These factors are categorized into four broad themes: computer (digital) literacy and attitudes, learner characteristics, training design and planning, and the learning environment along with organizational factors. Additionally, the study maps these factors to the intended outcomes of training (knowledge, skills, and attitude) and the participants engaged in the training sessions, providing a comprehensive overview of ongoing research and identifying areas for future investigation. Drawing insights from an evaluation of existing literature, the research not only points toward potential research directions but also offers practical implications for designing more effective online training sessions.

KEYWORDS

transfer of computer-based learning, transfer of e-learning, transfer of mobile learning, transfer of online education, transfer of online learning

1 | INTRODUCTION

Continuous upskilling has become imperative for companies to thrive in today's knowledge economy. Consequently, organizations are dedicating significant resources to a diverse range of training activities (Baldwin et al., 2009; Caporarello et al., 2019; Choi et al., 2021; Dolezalek, 2005). However, despite these investments, concerns persist within organizations regarding the transferability of training to the workplace (Baldwin & Ford, 1988; Garavan et al., 2019; Hutchins et al., 2010). The provision of training through novel mediums, such as online platforms, is likely to intensify these concerns due to limited insights into the factors influencing transferability within this domain. Even though the movement toward online training was expedited by technological advancements and the Covid-19 pandemic-related restrictions (Akdere et al., 2023; Coman et al., 2020; Dhawan, 2020; Mulder, 2021; Sobaih et al., 2020), the benefits and efficiencies of online training over face-to-face methods still persist in the post-pandemic era (Patino & Naffi, 2023). It is now essential for HRD professionals to identify and categorize the factors affecting the transferability of online training, enabling the understanding, differentiation, and adoption of best practices.

Effectively ensuring training transfer necessitates meticulous planning and strategizing (Machin, 2002; Yamnill & McLean, 2001). This, in turn, calls for an understanding of the factors influencing training transfer. Despite a growing emphasis on online training transfer, there remains a dearth of comprehensive studies that systematically synthesize the existing literature and discern the factors that facilitate successful online training transfer. This study aims to bridge this gap by proposing a taxonomy of factors based on a comprehensive synthesis of existing research. This taxonomy would serve as a guide for future research, aiding researchers in identifying potential new factors based on their relation to existing factors or their alignment with the broader categories identified in this study. It would also provide an indispensable checklist for HRD practitioners seeking to enhance online training transfer in real-world organizations.

By aligning the factors affecting training transfer with the purpose of training (knowledge, skills, and attitude) and the profile of trainees (industry types), this paper assists HRD professionals in designing effective online training programs tailored to the content of training and the intended audience.

The specific research questions addressed in this study are: RQ1: How can the existing literature on factors influencing the transfer of online training be systematically categorized into a coherent framework? RQ2: How can these factors be employed to provide a comprehensive overview of the current state of literature and to propose directions for future research?

To address these questions, this study employs a systematic literature review (SLR). SLR is a recommended method for summarizing and organizing existing literature in a specific area (Makarem & Wang, 2020). SLR is well-suited for addressing targeted research questions that necessitate content synthesis and comparison across articles (Snyder, 2019). Given that this article seeks to synthesize existing research in the evolving field of online training transfer for specific goals (constructing a taxonomy of factors), SLR is a suitable methodology. The SLR encompassed articles published between April 2001 and May 2021, a timeframe chosen based on the emergence of online training phenomena in the early 2000s.

Initial keyword-based exploration uncovered pertinent articles alongside unrelated subjects such as technology and material training transfers. To ensure accuracy and a systematic approach, PRISMA guidelines were adhered to. These guidelines facilitate a methodical and rigorous assessment of the current literature (Page et al., 2021). PRISMA is a widely used methodology for SLRs, and its application has demonstrated enhanced reporting quality

and methodological rigor in medical research (Panic et al., 2013). PRISMA guidelines have also found widespread application in SLRs within management research (Lim & Rasul, 2022). The initial search yielded 502 articles, out of which 33 were selected for review according to PRISMA guidelines. These articles were analyzed to develop a taxonomy of factors influencing the transfer of training in the online medium. The study identified four primary categories fostering training transfer in the online medium: digital literacy and attitudes, learner characteristics, instructional design, and learning environment and organizational support.

2 | TRANSFER OF TRAINING AND ONLINE MODE

Transfer of training refers to the degree to which the knowledge, skills, and attitudes acquired in a training program are effectively applied to enhance job performance (Laker & Powell, 2011, p. 112). Improving job performance necessitates the practical utilization of these KSAs (knowledge, skills, and attitudes) in real-world scenarios (Grossman & Salas, 2011; Schoeb et al., 2021). Therefore, the transfer of training assesses the extent to which trainees can apply the acquired KSAs from training to their job roles.

Research on training transfer has substantially expanded in recent times. However, discussions persist around defining, measuring, and contextualizing training transfer. A consensus definition and universally accepted measurement scale for training transfer have not emerged (Schoeb et al., 2021). This conceptual and operational uncertainty results in varied definitions and measurement tools employed by different researchers, leading to limited generalizability across studies. The implications of the lack of consensus and measurement tools are even more pronounced within digital media due to the diverse nature of online environments. For instance, digital devices differ in terms of size, processing power, and features, potentially impacting the transferability of digitally received training.

The rapid surge in online training due to technological advancements and Covid-19-driven restrictions (Coman et al., 2020; Dhawan, 2020; Sobaih et al., 2020) has given rise to diverse conceptual understandings of online training. Beyond conceptual and operational challenges, debates also center around the impact of different media on training effectiveness. While certain studies point out transfer-related distinctions between online and offline media (Kimball & Holyoak, 2000), meta-analyses have struggled to confirm these findings. For instance, Kaplan et al. (2021) found no significant differences in transfer effectiveness between extended-reality (XR) training and traditional mediums, except for physical tasks. This suggests that simulations in extended reality might be more effective for physical skill training.

The perceived influence of the training's goal on its effectiveness has also been explored. Blume et al. (2010) found that transfer magnitude is higher for open skills (focused on understanding and internalization) compared to closed skills (emphasizing specific behaviors). While research has not explicitly linked online training to better acquisition of closed skills (Kaplan et al., 2021), this could be due to inconsistencies in reporting training transfer studies (Blume et al., 2010) and the lack of control for contextual variables (Kaplan et al., 2021), warranting further sophisticated meta-analysis.

A related understudied area is the impact of the purpose of training, such as the transfer of knowledge, skills, and attitude, on transfer effectiveness. The author(s) failed to identify studies in this domain. The lack of granular reporting on the content of training, whose transfer was studied, has been identified by existing SLRs (Schoeb et al., 2021). Besides the lack of granularity, the SLR by Schoeb et al. (2021) also highlighted the overall absence of reporting on parameters in studies exploring the transfer of training. However, some studies have explored context-specific factors promoting transferability. For leadership training, trainer characteristics (Richter & Kauffeld, 2020) and evidence-based pedagogy (Seeg et al., 2022) have proven effective. Regarding the transfer of technical training, content and supervisor support have been found to play significant roles in training transfer (Richter & Kauffeld, 2020). Concerning entrepreneurship training, demographic factors and prior experience with such training are important for training transfer (Minja et al., 2022).

The differential impact of industry type on the transfer of training is also being investigated. Simosi (2012) examined the influence of self-efficacy on training transfer in the service industry, and attitudes on training transfer on transferability in the oil and gas industry. Yaqub et al. (2021) and Kodwani (2017) identified the positive impact of self-efficacy and organizational factors, respectively, on training transfer in public sector companies. Scholars are also identifying new factors that affect the transfer of training, such as team-based learning (Hart et al., 2019), culture (Brion, 2020; Gil et al., 2021), trainer conduct (Creon & Schermuly, 2022), job resources and demands (Salamon et al., 2022), learner readiness (Ghosh, 2022), implementation intention (Greenan, 2023), and the evaluation of training (Arabi & Garza, 2023).

Studies have attempted to identify the relative contribution of specific environmental conditions to training transfer (Hughes et al., 2020) and have employed more sophisticated statistical techniques. Studies are increasingly using sequential mediation techniques (Andoh et al., 2023; Blume et al., 2022; Kim et al., 2019), leading to a better understanding of the mechanisms. These novel factors generate substantial discussion and offer potential avenues for future research in the field of training transfer. Longitudinal studies are also being utilized, enhancing the validity of suggested causal relationships (Deckers et al., 2022).

In conclusion, Cheng and Hampson's (2008) clarion call for research on training transfer still remains valid. Although some research has been conducted to address the concerns they voiced, substantial issues in clarifying and measuring the construct and addressing questions regarding the impact of context, such as the medium and content of training, still persist. This study focuses on gaining an understanding of training in the online medium by identifying and classifying the factors that lead to transfer effectiveness in online training. This would enable a comparison of differences and similarities with research advancements in the more established domain of offline/face-to-face training (Clark, 2020; Fauth & González-Martínez, 2021a, 2021b; Galoyan et al., 2021; Martins et al., 2019; Stefaniak, 2020). The study also classifies earlier studies in terms of the purpose of training (knowledge, skills, and attitude) and the industry/domain of trainees. This classification can assist future researchers in conceptually linking factors to training content and participant profiles, driving specific and high-impact empirical research in the field of online training. HRD professionals can also use this classification to develop more customized training programs.

3 | METHODOLOGY

3.1 | Systematic literature review

Literature reviews serve as essential tools for synthesizing existing knowledge within a domain and identifying gaps (Xiao & Watson, 2019). They play a crucial role in the advancement of research (Borrego et al., 2014) by providing an avenue to amalgamate and reflect upon prior research, thus establishing a robust foundation for knowledge progression (Paré et al., 2015). In instances where a literature review lacks rigor and comprehensiveness, doubts may arise regarding the validity of its conclusions. As a result, scholars advocate for a systematic and thorough approach in the literature review process to address concerns pertaining to reliability, validity, and replicability. To this end, this study employs a systematic literature review (SLR) to synthesize existing knowledge concerning the transfer of online training.

3.1.1 | Research design

Conducting SLRs necessitates the adoption of standard frameworks to guide the analysis. An initial examination of databases using specific search terms revealed a substantial body of literature covering diverse subject matters sharing common keywords. Consequently, the need for a methodical approach, guided by clear exclusion and inclusion criteria, was emphasized. This study employs the Preferred Reporting Items for Systematic Reviews and

Meta-Analyses (PRISMA) guidelines. These guidelines were designed to assist scholars in methodically and meticulously analyzing literature within a domain, facilitating an extensive, thorough, and precise assessment (Page et al., 2021). PRISMA guidelines have found wide application in conducting SLRs within the management domain. Prior researchers have also employed PRISMA guidelines to synthesize research within the training domain (e.g., Kotera et al., 2019; MacLean et al., 2023). While alternative frameworks exist for conducting SLRs, there is no, to the best of the author(s) knowledge, study quantifying the quality of outcomes resulting from the use of different frameworks. Additionally, no evidence was found of the use of alternative frameworks for conducting SLRs within the training domain. Given the extensive adoption of PRISMA and the absence of compelling reasons to explore other frameworks, this study opts to use PRISMA guidelines.

3.1.2 | Search strategy

The search encompassed articles from three databases: Google Scholar, Web of Science, and EBSCO. The search employed the following keywords: Transfer of e-learning, Transfer of online learning, transfer of computer-based learning, transfer of mobile learning, and transfer of online education. The search scope was limited to the titles of the articles. The keywords were deliberately not enclosed in quotation marks to ensure the inclusion of the maximum possible number of papers. Consequently, if any portion of a keyword such as “online” or “mobile” appeared in the title, the paper was deemed suitable for further evaluation. The selection of all three databases utilized in the study was driven by their substantial article repositories, and their selection was motivated by their size and accessibility to the authors (refer to Table 1).

3.1.3 | Inclusion and exclusion criteria

The inclusion and exclusion criteria were developed initially, after which the authors engaged in a two-stage process. In the first stage, the articles were sorted based on abstract reading, followed by further scrutiny of the full text in the second stage. All authors participated in discussions and finalized the articles, first after assessing the abstracts and then after reviewing the full text, reaching a consensus on the articles to be included and excluded (refer to Figure 1). Articles were included in the systematic literature review (SLR) if they met the following criteria: (a) The article focused on factors determining the transfer of training, and (b) The study was conducted in an online medium, which encompassed e-learning, online learning, computer-based learning, mobile learning, and online education. The SLR systematically excluded studies that were: (a) Not relevant to the transfer of training or learning, (b) Not conducted in an online medium, (c) Did not primarily focus on the transfer of training despite containing the required

TABLE 1 Search and inclusion of articles in SLR (created by Author).

Row labels	EBSCO	Google Scholar	Web of Science	Grand total
Transfer computer-based training	3	13	4	20
Transfer e-learning	18	124	13	155
Transfer mobile learning	4	51	10	65
Transfer online education	12	11	3	26
Transfer online learning	22	142	35	199
Transfer online teaching	1	8	3	12
Transfer online training	4	15	6	25
Grand total	64	364	74	502

keywords, (d) Conference papers, and (e) Unpublished studies. In the initial step, a total of 502 articles were identified. Following the predetermined inclusion and exclusion criteria, 387 articles were excluded after abstract reading. A higher exclusion rate was observed due to many studies not aligning with the review's objective or inclusion criteria. For example, studies like "Transfer and maintenance effects of online working-memory training in normal ageing and mild cognitive impairment" by Vermeij et al. (2016) and "Promotion of e-Learning Solutions via Information Technology Transfer Concept and Baltic Regional Network" by Novickis et al. (2011) were excluded. During the review, it became apparent that some articles were excluded because the search strategy did not involve using the exact phrase "transfer of online training" within quotation marks. This strategy was chosen to encompass all relevant articles on the subject and prevent any potential omissions. However, this unintentionally resulted in the inclusion of numerous irrelevant articles discussing technology transfer in training or material training. As a result, a significant number of articles were excluded from analysis. Articles were excluded if they did not primarily focus on the transfer of online training or if they did not explicitly identify factors impacting online training. While many articles included the term "transfer" and pertained to training, they were centered on technology transfer or other forms of transfer, not training transfer. Consequently, these articles were excluded as they did not align with the study's objective. In the second stage, all 91 shortlisted articles were thoroughly read and evaluated based on the predetermined inclusion and exclusion criteria. Subsequently, 58 articles that did not meet the inclusion criteria were removed. All excluded articles discussed the transfer of online training, but they did not provide information on factors impacting online training transfer. Some of these articles presented findings in different contexts that did not align with the inclusion criteria of this study. For example, the article "Bi-directional online transfer learning: a framework" by McKay et al. (2020) was excluded as it did not align with the study's objective and inclusion criteria. The authors conducted this process concurrently in both stages and reached a consensus decision. Further details of this process are outlined in Figure 1. Ultimately, 33 articles were selected for systematic review.

4 | RESULTS AND FINDINGS

4.1 | Developing a taxonomy of factors enabling transfer of online training

4.1.1 | Identification of factors

A total of 33 articles were shortlisted for the review. Among these articles, 18 reported the results of quantitative studies, and 6 described qualitative studies. Eight articles presented literature reviews, while one article detailed a mixed-method study that employed both qualitative and quantitative techniques. For the quantitative studies, factors were identified by analyzing hypotheses and results. Factors hypothesized to facilitate the transfer of online training were considered as such if they were empirically supported by the hypotheses.

Various factors were identified from quantitative studies, including computer anxiety (Park & Wentling, 2007), organizational support, and supervisor support (Martins et al., 2019). In the case of the mixed-method study, factors were similarly identified from the text of the study. In qualitative studies, factors were identified based on the author(s)' use of qualitative techniques like interviews, focus groups, or anthropological studies. Attention was particularly given to sections discussing study results and findings. Factors identified from qualitative studies encompassed the design of training (Fauth & González-Martínez, 2021b) and teaching pedagogy (Galoyan et al., 2021).

Literature reviews on related topics provided another valuable source for identifying factors affecting the transfer of online training. Close reading of these reviews led to the identification of additional factors, such as technology readiness, infrastructure, organizational culture, instructional decisions, peer support, training content, communication between learners and trainers, opportunities for use, content delivery style, perceived utility, computer literacy barriers, networking objectives, and the need for networking (Clark, 2020; David, 2013; Fauth & González-Martínez, 2021a; Huffaker & Calvert, 2003; Kohn et al., 2010; Lim, 2012; Stefaniak, 2020).

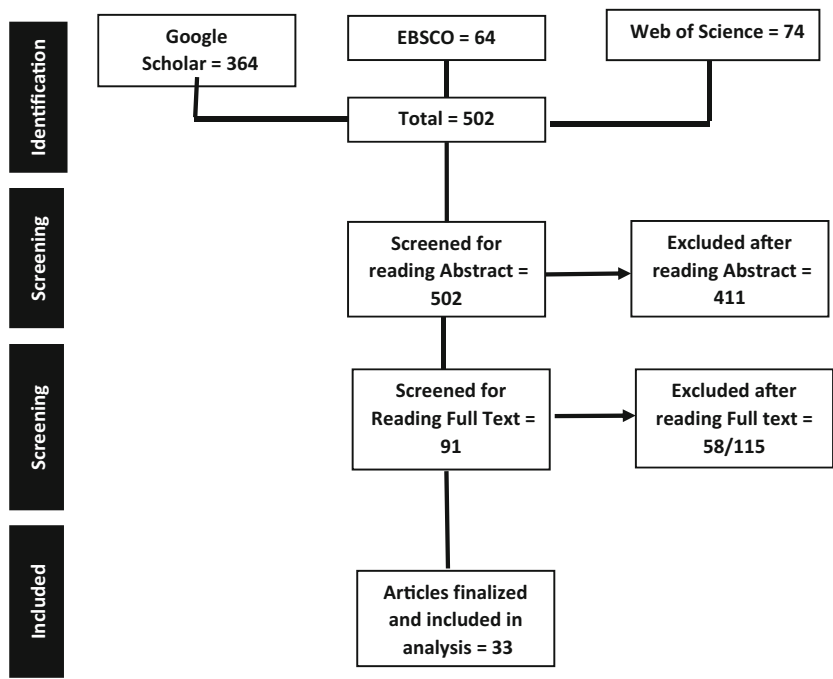


FIGURE 1 Inclusion or exclusion as per PRISMA guidelines (created by Author). The SLR systematically excluded studies: (a) not related to the transfer of training or learning; (b) not conducted in an online medium; (c) studies not focusing on the transfer of training even when they contained the requisite keywords; (d) conference papers; and (e) unpublished studies.

In total, 33 distinct factors were identified from the existing literature review. Detailed information about these factors can be found in Table 2. Some factors were mentioned across multiple studies. Table 3 provides an overview of the studies where each factor was identified, along with their mapping to the purpose and participant profile of the training where applicable. Additionally, Table 3 categorizes each factor into broader categories. The process used to categorize factors into these broader categories is outlined in the following section.

4.1.2 | Categorization of taxonomy of factors (taxonomy)

Several studies have developed taxonomies of factors that influence the transfer of training in face-to-face mediums (Baldwin & Ford, 1988; Burke & Hutchins, 2008; Lim & Morris, 2006; Park & Kim, 2016). However, as far as our knowledge goes, similar taxonomies have not yet been established for online training. The creation of a taxonomy for factors affecting online training would facilitate a comparison between factors influencing online and offline training. This comparison could also help pinpoint future research directions in online training and empower practitioners to leverage the well-developed literature on the transfer of offline training when designing training programs. Consequently, it is logical to investigate the factors identified in the transfer of online training and derive a taxonomy based on this study to guide future research (as depicted in Tables 2 and 3).

To construct a taxonomy, the factors identified in the studies were initially listed individually. Subsequently, similar factors were grouped together. The grouping process involved reaching a consensus among the co-authors. Factors were categorized into a single group if all the authors concurred. For instance, various factors such as learner internal values, learner satisfaction, learner attitudes, self-efficacy, and learner control all pertained to learner

TABLE 2 Summary of studies on transfer of online training.

SN	Author	Year	Type of study	Geography	Factors studied which impact transfer of training
1	Lim (2012)	2012	Literature review	NA	Technology readiness, infrastructure, organizational culture, organizational capability, and organizational readiness, teaching models, learner variables, instructional decisions
2	David (2013)	2013	Systematic review	NA	Peer support, training content, communication between learner and trainer
3	Park and Wentling (2007)	2007	Quantitative	US	Computer anxiety, computer confidence, computer liking, computer usefulness, satisfaction, learnability
4	Martins et al. (2019)	2019	Quantitative	Brazil	Practical application, trainees' reactions to training, learning strategies, peer, and organizational support, supervisor support
5	Joo et al. (2011)	2011	Quantitative	Korea	Organizational support, learning flow, learners' satisfaction
6	Hansen (2008)	2008	Quantitative	US	Communication ownership of the course material, and sense of community
7	Peters et al. (2012)	2012	Quantitative	Belgium	Enjoyment, utility and difficulty, interaction opportunities
8	Pham and Huynh (2018)	2018	Quantitative and qualitative	Vietnam	E-mail interaction, easiness and usefulness and social presence
9	Ramli et al. (2010)	2010	Quantitative	Malaysia	Learner's perceived utility/value, learner's motivation, learning design, training content, support from supervisors and peers
10	Zia-ur-Rehman and Shahzadi (2014)	2014	Quantitative	Pakistan	Learning usefulness, Internal value, learning environment, motivation for skill transfer, and satisfaction
11	Barak et al. (2016)	2016	Quantitative	Israel	Self-regulation, cognitive skills
12	Fauth and González-Martínez (2021a)	2021	Literature review	NA	Needs analysis, training and learning design, experience, motivation, self-management, opportunity for use, and environmental resources
13	Fauth and González-Martínez (2021b)	2021	Qualitative	Spain	Design of training
14	Hoyt (2013)	2013	Quantitative	UK	Higher attitudes and higher perceived behavioral control, higher goal-setting structure
15	Gunawardena et al. (2010)	2010	Quantitative	US	Self-efficacy, collegial support
16	Park et al. (2010)	2010	Quantitative	South Korea	Computer anxiety and confidence, self-efficacy, satisfaction, learnability, and efficiency, usability, design of the e-learning course
17	Chee et al. (2018)	2018	Quantitative	Malaysia	Needs analysis, training and learning design, experience, motivation, self-management, feedback, opportunity for use, institutional recognition, and environmental resources

TABLE 2 (Continued)

SN	Author	Year	Type of study	Geography	Factors studied which impact transfer of training
18	Huffaker and Calvert (2003)	2003	Literature review	NA	Teaching methodology and interactions during session
19	Granger and Levine (2010)	2010	Quantitative	US	Learner control
20	Van Doorn and Van Doorn (2014)	2014	Literature review	NA	Learning needs, learning styles, institutional support needs, educational culture and computer technology
21	Herrington et al. (2009)	2009	Quantitative	Australia	Quality of the online modules, training support
22	Stefaniak (2020)	2020	Literature review	NA	Instructors' technology know-how, communication between the instructor and students, with community partners, and among students, content delivery style, perceived utility, social support
23	Galoyan et al. (2021)	2021	Qualitative	US	Teaching pedagogy, instructional approaches and strategies
24	Fraj and Mbarek (2017)	2017	Qualitative	Tunisia	Motivation to transfer, peer support, supervisor support, atmosphere of organization
25	Brion (2019)	2019	Qualitative	Ghana	Participation opportunities, collaboration with team members and motivation to transfer
26	Kohn et al. (2010)	2010	Literature review	NA	Infrastructure, internet and technology access, usability, interaction, and computer literacy barrier
27	Clark (2020)	2020	Literature review	NA	Objectives and need for networking
28	Ferri et al. (2020)	2020	Qualitative	Italy	Digital skills of learner and teacher, structured content, interactivity and motivation and social and cognitive presence, interaction
29	Ronen (2008)	2008	Book	US	Usability, computer self-efficacy, supervisor support
30	Hennebury (2007)	2007	Quantitative	Germany	Situational cues or organizational climate factors, (goal, social, task) and consequences (negative feedback, positive feedback, or no feedback)
31	Merkley and Nichols (2007)	2007	Quantitative	China	Design of the training program, organizational policies/practices
32	Everett (2010)	2010	Qualitative	US	Course design, evaluation and social interaction
33	Trombley (2004)	2004	Quantitative	US	Self-efficacy and training usefulness

TABLE 3 Identification of categories and themes.

Category	Factor	Study	Purpose of training	Industry
Computer literacy and attitudes	Computer anxiety	Park and Wentling (2007)	Skill training (hard skill and soft skill)	Construction and agro machinery
		Park et al. (2010)	Not mentioned	Technology
	Computer confidence	Park and Wentling (2007)	Skill training (hard skill and soft skill)	Construction and agro machinery
		Park et al. (2010)	Not mentioned	Technology
	Computer liking	Park and Wentling (2007)	Skill training (hard skill and soft skill)	Construction and agro machinery
		Kohn et al. (2010)	NA (literature review)	NA (literature review)
	Computer literacy and skills	Park and Wentling (2007)	Skill training (hard skill and soft skill)	Construction and agro machinery
	Computer usefulness			
Learner characteristics	Learner satisfaction	Joo et al. (2011)	Not mentioned	Electronics
		Zia-ur-Rehman and Shahzadi (2014)	Knowledge transfer (two subjects)	Education (university)
	Learner motivation	Park et al. (2010)	Not mentioned	Technology
		Pineda-Herrero et al. (2011)	Not mentioned	Various industries
		Ramli et al. (2010)	Knowledge transfer (students of courses)	Education (University)
	Motivation to transfer	Fauth and González-Martínez (2021a)	NA (literature review)	NA (literature review)
		Zia-ur-Rehman and Shahzadi (2014)	Knowledge transfer (two subjects)	Education (university)
		Chee et al. (2018)	Not mentioned	Oil and gas
	Self-regulation or management	Fraj and Mbarek (2017)	Not mentioned	Not mentioned
		Brion (2019)	Skills training (leadership)	Various industries
		Pineda-Herrero et al. (2011)	Not mentioned	Various industries
		Barak et al. (2016)	Knowledge transfer (subject course)	Education (university)
		Fauth and González-Martínez (2021a)	NA (literature review)	NA (literature review)
		Chee et al. (2018)	Not mentioned	Oil and gas

TABLE 3 (Continued)

Category	Factor	Study	Purpose of training	Industry
Learning environment and organizational support	Self-efficacy	Gunawardena et al. (2010)	Knowledge transfer (engineering course)	Education
		Park et al. (2010)	Not mentioned	Technology
		Ronen (2008)	Not mentioned	Global company
		Trombley (2004)	Knowledge and skills (orientation program)	Retail company
	Trainee's reaction	Pineda-Herrero et al. (2011)	Not mentioned	Various Industries
		Martins et al. (2019)	Skill training (operational efficiency)	Public bank
	Enjoyment	Peters et al. (2012)	Behavioral training (stress prevention)	Education (university)
	Learner control	Granger and Levine (2010)	Knowledge and skills transfer (PowerPoint 2007)	Education (university)
	Cognitive skills	Barak et al. (2016)	Knowledge transfer (subject course)	Education (university)
	Attitudes and behavior	Hoyt (2013)	Software skills (application)	Education (university)
		Fauth and González-Martínez (2021a)	NA (literature review)	NA (literature review)
	Learner variables	Lim (2012)	NA (literature review)	NA (literature review)
	Technology	Lim (2012)	NA (literature review)	NA (literature review)
		Van Doorn and Van Doorn (2014)	NA (literature review)	NA (literature review)
		Stefaniak (2020)	NA (literature review)	NA (literature review)
Infrastructure		Kohn et al. (2010)	NA (literature review)	NA (literature review)
		Lim (2012)	NA (literature review)	NA (literature review)
		Kohn et al. (2010)	NA (literature review)	NA (literature review)
		Lim (2012)	NA (literature review)	NA (literature review)
	Organizational culture	Van Doorn and Van Doorn (2014)	NA (literature review)	NA (literature review)

(Continues)

TABLE 3 (Continued)

Category	Factor	Study	Purpose of training	Industry
Peer support		Fraj and Mbarek (2017)	Not mentioned	Not mentioned
		David (2013)	NA (literature review)	NA (literature review)
		Martins et al. (2019)	Skill training (operational efficiency)	Public bank
		Ramli et al. (2010)	Knowledge transfer (students of courses)	Education (university)
Communication/interaction		Gunawardena et al. (2010)	Knowledge transfer (engineering course)	Education
		Stefaniak (2020)	NA (literature review)	NA (literature review)
		Fraj and Mbarek (2017)	Not mentioned	Not mentioned
		David (2013)	NA (literature review)	NA (literature review)
		Hansen (2008)	Knowledge transfer (marketing subject)	Education (university)
		Brion (2019)	Skills training (leadership)	Various industries
		Clark (2020)	NA (literature review)	NA (literature review)
		Peters et al. (2012)	Behavioral training (stress prevention)	Education (university)
		Huffaker and Calvert (2003)	NA (literature review)	NA (literature review)
		Stefaniak (2020)	NA (literature review)	NA (literature review)
Organizational support		Kohn et al. (2010)	NA (literature review)	NA (literature review)
		Everett (2010)	Knowledge and skills training	Education (university)
		Hennebury (2007)	IT knowledge and skills (web security)	Financial services
		Martins et al. (2019)	Skill training (operational efficiency)	Public bank
		Joo et al. (2011)	Not mentioned	Electronics
		Herrington et al. (2009)	Knowledge transfer (for teachers)	Education

TABLE 3 (Continued)

Category	Factor	Study	Purpose of training	Industry
Online learning design and planning	Learning environment	Martins et al. (2019)	Skill training (operational efficiency)	Public bank
		Ramli et al. (2010)	Knowledge transfer (students of courses)	Education (university)
		Fraj and Mbarek (2017)	Not mentioned	Not mentioned
		Ronen (2008)	Not mentioned	Global company
		Pineda-Herrero et al. (2011)	Not mentioned	Various industries
	Opportunity to use	Zia-ur-Rehman and Shahzadi (2014)	Knowledge transfer (two subjects)	Education (university)
		Fauth and González-Martínez (2021a)	NA (literature review)	NA (literature review)
		Chee et al. (2018)	Not mentioned	Oil and gas
	Organizational climate	Fauth and González-Martínez (2021a)	NA (literature review)	NA (literature review)
		Chee et al. (2018)	Not mentioned	Oil and gas
		Pineda-Herrero et al. (2011)	Not mentioned	Various industries
	Organizational policies	Hennebury (2007)	IT knowledge and skills (web security)	Financial services
		Merkley and Nichols (2007)	Knowledge and skills training (competency)	Government agency
	Organizational readiness	Lim (2012)	NA (literature review)	NA (literature review)
Learnability	Training content	David (2013)	NA (literature review)	NA (literature review)
		Hansen (2008)	Knowledge transfer (marketing subject)	Education (university)
	Learnability	Herrington et al. (2009)	Knowledge transfer (for teachers)	Education
		Park and Wentling (2007)	Skill training (hard skill and soft skill)	Construction and agro machinery
		Park et al. (2010)	Not mentioned	Technology
	Application or usefulness	Stefaniak (2020)	NA (literature review)	NA (literature review)
		Ronen (2008)	Not mentioned	Global company

(Continues)

TABLE 3 (Continued)

Category	Factor	Study	Purpose of training	Industry
Learning strategy and design		Martins et al. (2019)	Skill training (operational efficiency)	Public bank
		Peters et al. (2012)	Behavioral training (stress prevention)	Education (university)
		Ramli et al. (2010)	Knowledge transfer (students of courses)	Education (university)
		Zia-ur-Rehman and Shahzadi (2014)	Knowledge transfer (two subjects)	Education (university)
		Trombley (2004)	Knowledge and skills (orientation program)	Retail company
		Park et al. (2010)	Not mentioned	Technology
		Kohn et al. (2010)	NA (literature review)	NA (literature review)
		Martins et al. (2019)	Skill training (operational efficiency)	Public bank
		Ramli et al. (2010)	Knowledge transfer (students of courses)	Education (University)
		Fauth and González-Martínez (2021a)	NA (literature review)	NA (literature review)
		Fauth and González-Martínez (2021b)	Skills training (for teachers)	Education (university)
		Park et al. (2010)	Not mentioned	Technology
		Chee et al. (2018)	Not mentioned	Oil and gas
		Galoyan et al. (2021)	Knowledge transfer (online degree program)	Education
		Everett (2010)	Knowledge and skills training	Education (university)
Easiness or Difficulty		Merkley and Nichols (2007)	Knowledge and skills training (competency)	Government agency
		Peters et al. (2012)	Behavioral training (stress prevention)	Education (university)
		Pham and Huynh (2018)	Knowledge transfer (mentioned)	Education (university)

TABLE 3 (Continued)

Category	Factor	Study	Purpose of training	Industry
Need Analysis		Ramli et al. (2010)	Knowledge transfer (students of courses)	Education (university)
		Fauth and González-Martínez (2021a, 2021b)	NA (literature review)	NA (literature review)
		Chee et al. (2018)	Not mentioned	Oil and gas
		Van Doorn and Van Doorn (2014)	NA (literature review)	NA (literature review)
Goal Setting		Hoyt (2013)	Software skills (application)	Education (university)
		Hennebury (2007)	IT knowledge and skills (web security)	Financial services
Learning Flow		Joo et al. (2011)	Not mentioned	Electronics
		Lim (2012)	NA (literature review)	NA (literature review)
Teaching Models		Huffaker and Calvert (2003)	NA (literature review)	NA (literature review)
		Galoyan et al. (2021)	Knowledge transfer (online degree program)	Education
Teaching Methodology		Stefaniak (2020)	NA (literature review)	NA (literature review)
		Everett (2010)	Knowledge and Skills Training	Education (University)
Teaching Pedagogy		Lim (2012)	NA (literature review)	NA (literature review)
Content Delivery				
Evaluation				
Instructional Decisions				

attributes. Therefore, they were collectively placed under the category titled “learner characteristics.” Another set of factors revolved around the learner’s digital skills and attitudes toward digital devices, encompassing aspects like computer anxiety, computer awareness, computer liking, and computer confidence. These were consolidated into a category termed “computer (digital) literacy and attitudes.” Similarly, other categories like “online learning design and planning” as well as “learning environment and organizational support” were devised using this strategy.

4.1.3 | Research trends across the four categories

The research conducted within the established taxonomy of transfer of online training, encompassing categories such as computer literacy and attitudes, learner characteristics, online learning design and planning, and learning environment and organizational support, is further examined to assess its presence and growth across different time periods since April 2001, as reflected in the literature incorporated in this study.

The temporal trends of research are outlined in Table 4. The earliest paper included in this systematic literature review (SLR) was published in 2003. During the period from 2001 to 2005, only two relevant articles were published. Research activity saw an increase during the 2005–2010 period, often referred to as the Web 2.0 era, but experienced a decline in the subsequent 2011–2015 period. However, there has been a resurgence in research output during the 2016–2021 period. Given the escalating sophistication of technological systems and the acceleration of online migration induced by the pandemic, the authors anticipate a growing trend of research in the foreseeable future.

Figure 2 illustrates the temporal evolution of research focus within different categories of the transfer of online training. In the initial period spanning from 2003 to 2007, considerable emphasis was placed on computer literacy

TABLE 4 SLR articles shortlisted (created by Author).

Period	Number of papers
2001–2005	2
2006–2010	13
2011–2015	7
2016–2021	12

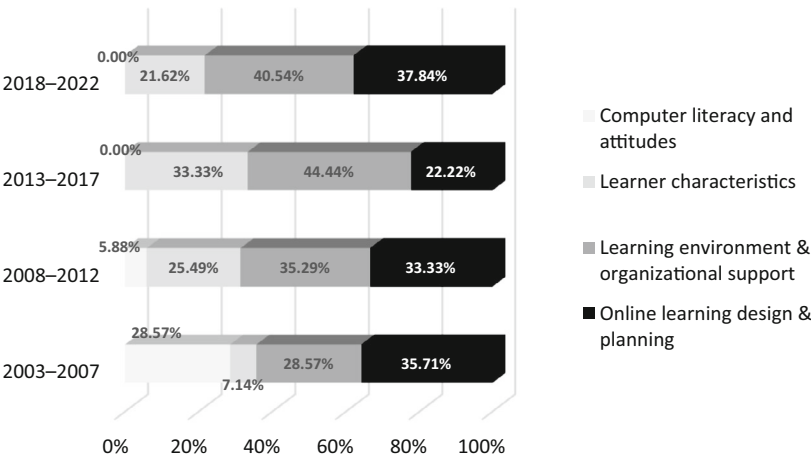


FIGURE 2 Available articles on the factors in the last 20 years.

and attitudes, alongside the exploration of other categories. This emphasis is understandable, given that this period marked the early stages of widespread computer usage in many parts of the world.

Subsequently, research in the realm of transfer of online training has witnessed growth across the other three categories. Starting from 2018, there has been a decline in research assessing computer literacy and attitudes, accompanied by an upsurge in studies centered around the remaining three categories. Notably, the most substantial concentration has been on learning design and planning, as well as learning environment and organizational support. Conversely, studies pertaining to the category of learner characteristics were relatively minimal during the initial years from 2003 to 2007 but have shown an increasing trend in the subsequent years.

As shown in Table 3, the category of Learner Characteristics has garnered extensive attention, with nearly 14 factors investigated across 22 studies. This prevalence may indeed be the highest among the four categories within this taxonomy of factors impacting the transfer of online training. Similarly, within the Computer Literacy and Attitudes category, 7 factors have been explored across 6 prominent studies.

Furthermore, upon closer examination of the factors, it becomes evident that certain elements have been the subject of a substantial number of studies (as indicated in Table 3). For instance, Communication and Interaction have been scrutinized in 14 studies, Learning Strategy and Design in 10 studies, Application and Usefulness in 12 studies, and Organizational Support in 10 studies, among others. However, some crucial factors have only been investigated in a limited number of studies, with examples including Computer Anxiety, Organizational Climate and Policies, Cognitive Skills, Content Delivery, and Instructional Decision, as detailed in Table 3. These factors, despite having fewer associated studies, hold potential as avenues for further future research.

The broad categories identified in Table 3, along with the individual-level factors, are visually represented in Figure 3. In the subsequent subsections, each of the categories is described in detail.

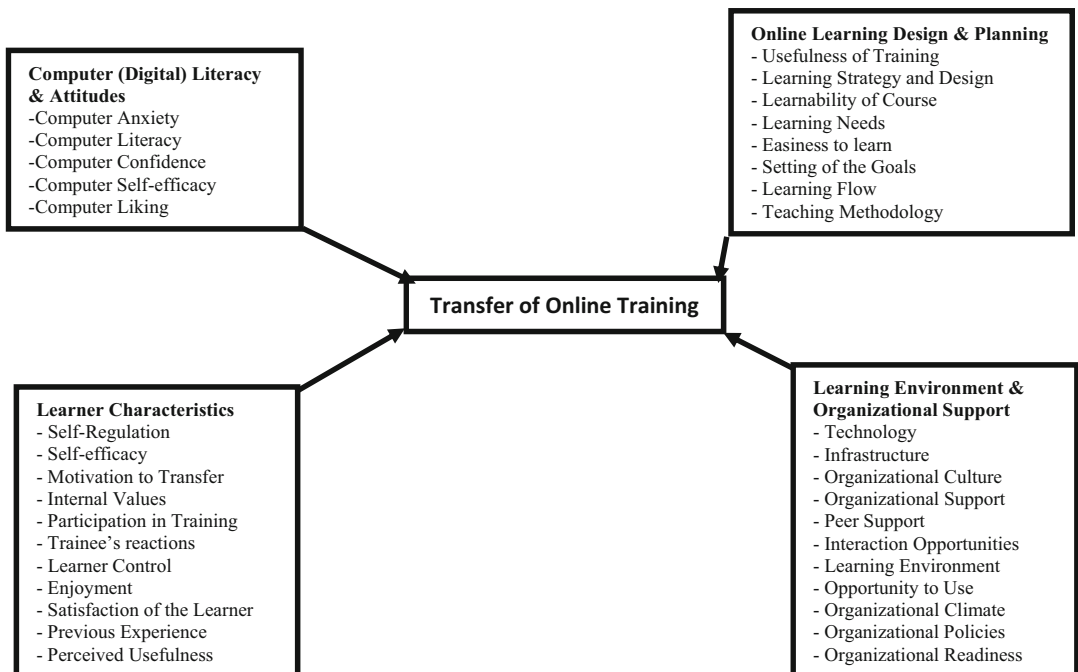


FIGURE 3 Categorization of factors affecting the transfer of online training.

4.2 | Description of the categories

4.2.1 | Computer (digital) literacy and attitudes

Several articles (e.g., Lim, 2012; Pham & Huynh, 2018; Ronen, 2008) have explored the positive influence of learner competencies in effectively using digital devices and exhibiting favorable attitudes toward digital mediums in facilitating the transfer of online training. Building on this insight, the factor of digital literacy and attitudes was classified as contributing to the successful transfer of online training. Digital literacy encompasses the knowledge and skills required for the efficient operation of digital devices, while digital attitudes encompass individuals' behaviors and attitudes toward digital devices and their usage, which can be either positive or negative.

Variables falling within this category include knowledge of technological advancements in the realm of digital devices (Clark, 2020; Jin et al., 2021; Kohn et al., 2010), strong computer confidence, perceived usefulness of computers, affinity for computer usage, and computer anxiety levels (Park et al., 2010; Park & Wentling, 2007). Similarly, this category encompasses attitudes specific to the digital realm, such as computer self-efficacy and low computer anxiety (Pham & Huynh, 2018; Ronen, 2008). The factors observed within this category are distinctive to the transfer of online training and serve as differentiating factors from traditional face-to-face training. The importance of positive attitudes toward computing devices for successful online training suggests the need to develop confidence and self-efficacy of learners in a digital environment. One way to achieve the same is by improving usability of the human-computer interface, especially for older learners (Chou & Hsiao, 2007).

4.2.2 | Learner characteristics

Learner characteristics have been recognized as relevant in the context of face-to-face training (Sookhai & Budworth, 2010), and the findings from this systematic review of studies (e.g., Huffaker & Calvert, 2003; Ramli et al., 2010; Ronen, 2008) further suggest that learner-specific attributes play a crucial role in determining the effectiveness of online training transfer. Thus, learner characteristics have been identified as a factor influencing the transfer of online training. Learner characteristics refer to stable behavioral patterns and general attitudes toward the learning process and pedagogy, which impact overall learning outcomes.

Various characteristics of learners contribute to the effective transfer of training, including self-regulation (Barak et al., 2016; Fauth & González-Martínez, 2021a, 2021b), self-efficacy (Gunawardena et al., 2010; Park et al., 2010; Trombley, 2004), motivation for transfer (Chee et al., 2018; Zia-ur-Rehman & Shahzadi, 2014), motivation to learn (Fauth & González-Martínez, 2021a, 2021b; Ramli et al., 2010), internal values (Zia-ur-Rehman & Shahzadi, 2014), and active engagement in training (Granger & Levine, 2010). Additional factors such as trainee's reactions (Martins et al., 2019), learner control (Granger & Levine, 2010), enjoyment (Peters et al., 2012), learner satisfaction with training (Pineda-Herrero et al., 2011), cognitive skills (Barak et al., 2016), and perceived usefulness of online training (All et al., 2016; Fauth & González-Martínez, 2021a, 2021b; Kohn et al., 2010; Trombley, 2004) also hold significance for the effective transfer of online training.

4.2.3 | Training design and planning

The review of the selected articles (e.g., Chee et al., 2018; Fauth & González-Martínez, 2021a, 2021b; Lim, 2012) suggests that the proper development and delivery of instructional material contribute to the effective transfer of online training. As a result, instructional design, defined as the process of systematically creating, developing, and delivering instructional materials and experiences—both digital and face-to-face—was identified as a factor

influencing the transfer of online training. The goal of instructional design is to enhance learning efficiency, effectiveness, engagement, appeal, and inspiration (Merrill et al., 1996).

Key variables that promote the effective transfer of online training and are categorized under instructional design include training content (David, 2013; Hansen, 2008), course usefulness (Ramli et al., 2010; Ronen, 2008; Stefaniak, 2020), learning strategy and design (Everett, 2010; Fauth & González-Martínez, 2021a, 2021b; Galoyan et al., 2021; Martins et al., 2019), course learnability (Park & Wentling, 2007), ease of learning (Peters et al., 2012), need analysis (Van Doorn & Van Doorn, 2014), and goal setting (Hennebury, 2007). Other identified factors include learning flow (Joo et al., 2011), evaluations (Everett, 2010), teaching models (Lim, 2012), and teaching methodology and pedagogy (Galoyan et al., 2021; Huffaker & Calvert, 2003).

4.2.4 | Learning environment and organizational support

Organizational climate has already been found relevant in the transfer of face-to-face training (Cromwell & Kolb, 2004; Egan et al., 2004; Martin, 2010; Rouiller & Goldstein, 1993; Xiao, 1996), and this section aims to assess its relevance in the context of online training transfer. The physical, social, and cultural settings also impact the effectiveness of the transfer of online training (as seen in Lim, 2012; Van Doorn & Van Doorn, 2014; Zia-ur-Rehman & Shahzadi, 2014). These physical, social, and cultural settings, including teacher/trainer characteristics and organizational factors, define the learning environment and organizational factors. The learning environment and organizational factors are significant categories of variables that influence the transfer of online training.

Several variables have been identified under this category, including technology (Lim, 2012; Van Doorn & Van Doorn, 2014), infrastructure (Kohn et al., 2010), organizational culture (Fraj & Mbarek, 2017), peer support (David, 2013; Martins et al., 2019), interaction opportunities (Clark, 2020; Everett, 2010; Hansen, 2008), organizational support (Herrington et al., 2009; Joo et al., 2011; Ronen, 2008), learning environment (Zia-ur-Rehman & Shahzadi, 2014), opportunity to use (Fauth & González-Martínez, 2021a, 2021b), organizational climate (Hennebury, 2007), organizational policies (Merkley & Nichols, 2007), and organizational readiness (Lim, 2012).

4.3 | Exploring the current state of research on contextual parameters

The study's objectives involve a comprehensive exploration of the existing literature landscape and the identification of potential research directions (RQ2). Existing literature has identified the absence of reporting on parameters as a limitation in the current literature on training transfer. Hence, this study explores whether existing literature has (1) captured the contextual parameters of training and (2) measured the differential impact of any factor in differing contexts. Two contextual factors were identified as most relevant by the authors: purpose of training and purpose of training. Many studies did not mention any or both of the parameters. No study measured the impact of context on the potency of a factor. The details are mentioned in Table 3 and discussed in subsequent sections. This approach has the potential to provide further layers of insight that can aid in refining research focus, addressing current research gaps, and uncovering possible paths for future investigation.

4.3.1 | Aligning identified factors to purpose of training

The purpose of training is to enhance an employee's job-relevant knowledge, skills, and attitudes, as highlighted by Schoeb et al. (2021). Consequently, when investigating the factors responsible for the transfer of training, it is crucial to consider the training's intended objectives. In the conducted literature review, five studies focused on skill

transfer, covering diverse areas such as software, leadership, teaching, operational, and other hard and soft skills. These skill-focused training programs were conducted in both industry and university settings.

Furthermore, eight studies centered on knowledge transfer, all of which took place within educational institutes or universities. Additionally, five studies examined both knowledge and skills transfer. The remaining 15 studies were either literature reviews or lacked specific information about the type of training investigated.

For studies that examined training focused on knowledge, skills, and attitudes, various specific factors were analyzed, as indicated in Table 3. For instance, under the category of 'Computer Literacy and Attitudes,' factors like computer anxiety, liking, and confidence were mentioned in studies with a focus on skill transfer. On the other hand, the category 'Learner Characteristics,' encompassing factors such as satisfaction, motivation, self-regulation, and self-efficacy, was included in studies centered on knowledge transfer.

Notably, the majority of the studies in this review concentrated on training programs where knowledge was the primary emphasis across all the categorized areas. This observation suggests that there is a research gap concerning the transfer of online training, specifically in cases where the training's purpose is to develop skills or attitudes. Further investigation in these domains could provide valuable insights into effective training transfer.

4.3.2 | Aligning identified factors to profile of participants

The impact of various factors on training transfer is contingent upon the institutional setting, as evidenced by differential effects observed among participants from industry and educational institutions (Argote et al., 1990; Cree & Macaulay, 2000). Therefore, this study identifies the importance of considering the institutional context in analyzing the factors affecting online training. Furthermore, this research aims to discern the type of participants (i.e., from industry or educational institutes) involved in online training and examines the range of factors studied across the articles included in the literature review.

The literature review incorporates a total of 33 studies, among which 11 were conducted in industry settings, providing training to employees in various sectors such as oil and gas, financial services, electronics, technology, and public banking. Some studies did not specify a particular industry as they involved participants from diverse sectors. Additionally, 13 studies were conducted in the education sector, focusing on training programs within institutes or universities, where the participants were students. The remaining eight studies were literature reviews that encompassed a variety of different research.

A closer examination of the types of industries involved in the studies (as shown in Table 3) reveals that studies concentrating on knowledge acquisition predominantly took place in educational institutes and universities, with students as the participants. On the other hand, studies focusing on skill development were conducted with employees from diverse industries, such as oil and gas, construction, and banking.

Another noteworthy observation is that the first category, "computer liking and attitude," received more attention in industry settings, whereas the other three categories were studied in both industry and academic institutes. Despite the findings presented in Table 3, there remains ample room for further research on this subject in various other industries and regions around the world.

5 | DISCUSSION AND INTERPRETATION OF RESULTS

This study employed a rigorous and systematic review of existing literature spanning the last 20 years to propose a taxonomy of factors that can impact the transfer of online training. Forty-six factors influencing the transfer of training in the online medium were identified and grouped into four categories: computer literacy and attitudes; learner characteristics; online learning design and planning; and learning environment and organizational support. This comprehensive categorization can aid HRD professionals in developing checklists to better design and execute online

training programs. The synthesis of research findings can guide future researchers in identifying new factors and associating them with broader categories not explored in current literature. Specific organizational roles can also be designated to assume responsibility for distinct factors. Trainers could be tasked with overseeing learner strategy and training usefulness, while management might be responsible for technology, infrastructure, and organizational culture. The HR department and individual learners should take responsibility for factors like learner self-efficacy and other individual-level aspects.

Comparing this taxonomy with earlier ones based on the transfer of training in offline mediums revealed the novel category of computer literacy and attitude for the transfer of training in online contexts. Factors within this category include computer anxiety, computer confidence, and computer liking. In striving for effective training transfer, HRD practitioners should assess trainees based on these parameters. If deemed necessary, specialized sessions aimed at reducing computer anxiety and enhancing computer confidence and liking might need to be conducted.

The taxonomy of factors developed in this study includes many elements, such as self-efficacy, peer support, learning strategy, and training usefulness, that are common to both online and offline training transfer. Given these commonalities, researchers can enhance our understanding of training transfer in an online context by exploring factors known to impact offline training transfer but not yet studied in the online medium. Moreover, for a given factor, researchers can investigate its relative importance in face-to-face and online contexts. This research direction can serve as a foundation for further qualitative and quantitative studies on the subject, contributing to the expansion of the current knowledge base.

In addition to identifying commonalities in the transfer of online and offline training, the study underscores the dearth of context-specific studies in this domain. For more than half of the studies, the specific training purpose—knowledge transfer, skill acquisition, attitude change, or behavioral transformation—was not identified. The study correlated the training purpose with the identified factors wherever feasible. This correlation can facilitate future researchers in exploring the nuanced relationship between training purpose and specific factors. Understanding these micro-relationships would aid institutions in better planning their training sessions by pinpointing and prioritizing the most crucial factors. The study also linked specific factors to participant types, as explored in the existing literature. This linkage can help identify specific research gaps for future investigations and potentially determine the institution-specific significance of factors.

5.1 | Limitations and future research directions

The scope of the literature review was confined to three databases: Google Scholar, Web of Science, and EBSCO. Considering that all three databases are major repositories of scientific literature, the author(s) are confident that the literature search effectively captured a wide array of articles. However, it is important to acknowledge the possibility of overlooking relevant articles from other databases. Similarly, even though the author(s) meticulously selected keywords, the potential for missing pertinent keywords cannot be completely dismissed. Another limitation was the restriction of your keyword search to only the titles of articles, potentially leading to the omission of articles that lack keywords in their titles.

The study revealed that existing literature has employed various definitions and operationalizations of the central construct—transfer of training. However, very few studies have delved into examining the equivalence among different scales utilized. Consequently, establishing the convergent validity of these diverse measures presents an area ripe for future research. Subsequent investigations could also explore avenues for refining the existing scales (Schoeb et al., 2021).

Drawing on insights from the literature review, other potential areas for research have also come to light. One notable gap pertains to the dearth of research assessing the ease of transferring various types of training content. This absence is somewhat surprising since the ease of transferability should ideally wield considerable influence over the determination of training content. The study suggests employing the KSA framework to broadly classify training

content based on its purpose. Future studies can rank different training content in terms of their transferability ease and pinpoint specific factors facilitating efficient transfer for each kind of training. Additionally, future researchers could discern which factors hold greater importance in distinct institutional settings. A comprehensive and validated alignment of factors, institutional contexts, and training content (purpose) could aid in designing more effective training programs, ultimately leading to improved learning outcomes and enhanced ROI.

Another avenue for future research involves refining the methodology. Current research tends to treat all digital mediums similarly with regard to their potential for learner interaction. Subsequent investigations could explore the influence of the sophistication level of digital mediums on motivating training transfer. In terms of research design, longitudinal studies represent an untapped direction. The predominant use of cross-sectional studies in current research (Jin et al., 2021; Joo et al., 2011) constrains authors from establishing causal relationships.

6 | CONCLUSION

Online provision of training has expanded rapidly due to technological advancements and the impact of the Covid-19 pandemic. However, there has been no comprehensive overview of the factors that promote the transfer of training in the online medium. The study aimed to overcome the same by identifying and classifying the factors promoting the transfer of online training (RQ1) and using the identified factors to evaluate the existing literature and suggest new research directions (RQ2). To achieve the objectives, a SLR was performed.

Based on the SLR, 33 relevant articles were identified and analyzed. The analysis of the articles led to the identification of 42 factors, which were classified into four categories: computer literacy and attitudes; learner characteristics; learning environment and organizational support; and online learning design and planning. One notable contribution of the study is the identification of computer literacy and attitudes as a new category, specifically relevant to online training.

An analysis of current literature on these factors suggests that few factors such as usefulness of training; learning strategy and design and organizational support have been explored in multiple studies. On the other hand, relatively few studies have explored the factors categorized under computer literacy and attitudes. As this category is unique to online training, there is scope for future research in this area. The study also explored whether existing literature has examined the impact of contextual parameters, such as the purpose of training and participant profile, on the efficacy of the factors. The findings reveal that reporting on contextual parameters needs strengthening with many studies providing no information on either or both of the parameters. No study examined the possible moderating impact of the contextual parameters. Future studies are advised to report the contextual parameters and evaluate their impact on the efficacy of factors for gaining an improved understanding of the phenomenon.

6.1 | Practical implications

The advent of the information technology revolution has made online training an attractive option due to its extensive reach, cost-effectiveness, and convenience (Szymkowiak et al., 2021). Moreover, the ongoing Covid-19 pandemic has accelerated the adoption of online training. Considering that training constitutes a substantial investment for organizations, its effectiveness becomes a crucial concern (Almahasees et al., 2021; Paudel, 2021). Researchers have raised questions about the effectiveness of training transfer for online versus traditional mediums (Liu et al., 2019). The challenge of training transfer has been a longstanding issue, and it becomes even more critical for Human Resource Development (HRD) professionals as learning transitions to the online mode, where significant efforts, time, and resources could potentially be at stake (Hutchins et al., 2010).

This literature review delves into the factors that can influence the transfer of online training, encompassing both specific and analogous elements found in face-to-face environments. Noteworthy specific factors identified for

online training include computer literacy, computer self-efficacy, computer anxiety, computer confidence, liking, technology readiness of the organization, infrastructure, trainer's technical skills, online interaction, and online evaluation. These factors demand careful consideration while developing and delivering online training sessions. The study presents a taxonomy of factors, offering HRD professionals insights into various categories that can aid in comprehending the factors affecting effective transfer in online training.

Given that many corporates and institutes have already adopted online training and seek to enhance transfer outcomes, this study provides valuable insights that can inform their strategies. The findings enable organizations to assess the identified factors and point out the opportunities for improving online training effectiveness and transfer. Furthermore, this study contributes to existing knowledge by presenting factors exclusive to and more relevant to the online mode of training. These findings can serve as future research directions for scholars to explore further using qualitative or quantitative studies.

The study also associates the identified factors with the training content (purpose) and institutional setting, drawing from the extant literature. The KSA framework is utilized to classify training based on its content, indicating that modules in educational institutes emphasize knowledge transfer, while those in corporate settings focus more on skills transfer. However, the lack of specific details in many research papers regarding the type of training or industry hampers efforts to comprehensively map the current state of knowledge or identify precise research questions. The authors hope that this mapping will inspire further research to identify specific configurations of factors, content, and institutions that lead to improved online training transfer outcomes.

DATA AVAILABILITY STATEMENT

The data (studies) that support the findings of this systematic literature review are primarily available in various databases like Google Scholar, EBSCO and Web of Science (as being referenced in document). These data were derived from the resources available in the public domain.

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